

Mohammed Saferi Rahman

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Education

Florida State University

PhD in Computer Science (In Progress)

MS in Computer Science (Dec, 2024)

- Advisor: Dr. Te-Yen Wu • Lab Name: MakeX • Research Area: HCI, UbiComp, AI
- Current CGPA: 3.77

Tallahassee, FL, United States

2022 – 2027 (expected)

BRAC University

BS in Computer Science

- CGPA: 3.55
- Graduated with Distinction • Dean's List: Fall '19, Spring '19

Dhaka, Bangladesh

2016 – 2020

Technical & Personal Skills

- **Programming Languages:** Highly Proficient Python, C/C++, Java, JavaScript, LaTeX
- **Frameworks & Libraries:** Pytorch, TensorFlow, Scikit-learn, HuggingFace, Transformers, OpenCV, Keras, Matplotlib, Pandas, Numpy, React Native, Expo, Unity, Firebase
- **Database:** MongoDB, SQL Server, Apache Spark, MySQL, XAMPP
- **Tools & Platforms:** Linux, macOS, Windows, Git, Jupyter, VS Code, Arduino IDE

Publications

- ★ "Double Deep Q-Learning and Faster R-CNN-based Autonomous Vehicle Navigation and Obstacle Avoidance in Dynamic Environment," In *Sensors Journal*, 2021. ([DOI](#))
- "Escape Rooms in Nursing Education: How GPA, Collaborative Problem Solving, and Gameful Experience are related," In *Proceedings of the 18th International Conference of the Learning Sciences-ICLS*, 2024. ([DOI](#))
- "Reinforcement Learning based Autonomous Vehicle for Exploration and Exploitation of Undiscovered Track," In *proceeding of the International Conference on Information Networking (ICOIN), IEEE Xplore, Spain*, 2020. ([DOI](#))
- "Autonomous Surface Vehicle for Real-time Monitoring of Water Bodies in Bangladesh," In *proceeding of the Global OCEANS Conference, IEEE, Singapore - US Gulf*, 2020. ([DOI](#))

Research Experience

- **JDR-LLM: Inferring Job Demands and Resources of PhD Students Using Multimodal LLM-Based Framework**
 - Publication: Submitted to CHI'26 (Under Review). ([Preprint](#))
 - Developed a multimodal LLM framework to infer PhD students' weekly JDR scores by analyzing mobile sensor data and digital interaction data through structured two-stage prompting and personalized calibrations.
 - Conducted a four-week in-the-wild study with 14 PhD students, integrating custom smartphone sensing and institutional data pipelines to model perceived stress and academic support.
 - Achieved high alignment with self-reported well-being scores (e.g., MAE $\approx$ 0.35, Spearman $\rho \approx$ 0.88), demonstrating the feasibility of continuous, passive monitoring of PhD students' well-being using LLMs.
- **Emotion-Aware Virtual Patient System for high-fidelity clinical communication training ([codebase](#))**
 - Designed and implemented an LLM based, emotion-aware Virtual Patient (VP) system using Unity, featuring neural TTS, viseme-synchronized facial animations, and configurable affect dynamics to simulate realistic clinical scenarios.
 - Developed a multimodal perception stack combining transformer ASR, prosodic analysis (F0, energy, speaking rate), and vision features (MediaPipe/OpenCV, DeepFace) to infer latent interaction states for adaptive dialogue generation.
 - Integrated a persona-conditioned LLM dialogue policy with real-time coaching, safety constraints, and SOFTEN-based nonverbal feedback dashboards, enabling high-fidelity clinical communication training and planned comparative evaluations against actor-based baselines.
- **Capacitive-Sensing Garments (CSG) for recognizing Dyadic Social Interaction**
 - Designed and fabricated multi-channel capacitive-sensing garments using FDC2214 sensor arrays and custom switching networks to capture fine-grained torso and arm proximity signals during dyadic interactions.
 - Developed a real-time data acquisition pipeline with microcontroller firmware (MSP430/Arduino), dynamic electrode selection, and synchronized recording of CSG signals and overhead video for labeled interaction datasets.
 - Conducting structured dyadic interaction studies and applied signal processing and classification pipelines to recognize social interaction types, laying groundwork for embodied HCI and wearable sensing applications. (*Work in progress*)

- **Text2IMU: Synthesizing Inertial Sensor Signals from Natural-Language Activity Descriptions** ([codebase](#))
 - Developed a framework that generates realistic inertial sensor signals from natural-language activity descriptions, enabling data augmentation and rapid prototyping for human activity recognition (HAR).
 - Designed a prompt-engineered LLM pipeline to parse activity semantics and synthesize time-series IMU data, with configurable parameters for sampling rate, motion intensity, and temporal patterns.
 - Evaluated generated signals against real IMU recordings, demonstrating their utility for training and benchmarking activity recognition models without costly data collection.
- **Double Deep Q-Learning and Faster R-CNN-based Autonomous Vehicle Navigation and Obstacle Avoidance in Dynamic Environment**
 - Publication: Sensors Journal, 2021. ([DOI](#))
 - Developed an autonomous vehicle navigation framework that fuses Double Deep Q-Learning (DDQN) for decision-making with Faster R-CNN for real-time object detection and classification in dynamic environments.
 - Implemented the system using Keras and TensorFlow, trained object detectors on the Open Image V5 dataset, and optimized DDQN hyperparameters for lane changing, acceleration, and obstacle avoidance.
 - Validated the integrated model in a GTA V game engine simulator, achieving 94% object detection accuracy and demonstrating precise navigation and obstacle avoidance compared to baseline Q-learning methods.

Notable Projects

- **ParkNet: Context-Aware Travel Itinerary Generation from Real-Time Signals** ([codebase](#))
 - Designed a web-based platform to help travelers discover and plan personalized trips to U.S. national parks through flexible itineraries, real-time information, and conservation-focused content.
 - Conducted comprehensive user research to identify travel challenges and preferences, informing platform features such as budget-conscious recommendations, itinerary customization, and sustainability integration.
 - Developed mid-fidelity prototypes in Figma, illustrating key user flows including personalized suggestions, group itinerary accommodation, and dynamic travel updates.
- **Generalizing University Ranking System from websearch** ([codebase](#))
 - Developed a comprehensive university ranking system that integrates data from multiple global ranking sources (Unirank, Webometrics, QS, etc.) to create a unified dataset covering thousands of institutions worldwide.
 - Implemented data collection, parsing, and imputation pipelines using Python, web scraping, and collaborative filtering (e.g., NMF) to handle missing rankings and merge heterogeneous datasets into a single structured CSV.
 - Containerized the entire system with Docker for reproducible deployment, enabling users to run the ranking pipeline end-to-end and generate imputed and combined global ranking tables.
- **Autonomous Surface Vehicle for Real-time Monitoring of Water Bodies in Bangladesh**
 - Designed and built a GPS-guided autonomous surface vehicle equipped with pH, turbidity, and temperature sensors, enabling real-time water quality monitoring across multiple preset locations.
 - Integrated GSM and Wi-Fi communication modules (SIM900A, ESP8266) for live data transmission to cloud servers, along with a pumping system for on-board sample collection and lab analysis.
 - Deployed and tested the system in five major lakes in Dhaka, successfully collecting and analyzing spatial water quality data to assess pollution levels and environmental trends.

Relevant Courses

- **Graduate:** Human-Computer Interaction (HCI), Advanced Computer Vision, Introduction to Data Science, Advanced Data Mining, Analytic Methods, Data and Computer Communication, Concurrent, Parallel, and Distributed Programming, Theory of Automata and Formal Languages, Computer & Network System Administration.
- **Undergraduate:** Artificial Intelligence, Neural Networks, Algorithms, Data Structures, Operating Systems, Computer Architecture, Database Systems, Digital System Design, Software Engineering, Probability & Statistics, Linear Algebra, Differential Equations.

Awards & Accolades

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| ◦ Research Excellence Award 2021
<i>BRAC University, Dhaka</i>
Awarded by the VC | ◦ Runner Up - Project Showcasing 2017
<i>Islamic University of Technology, Dhaka</i>
IUT 9th ICT Fest |
| ◦ Best Poster Paper Award Dec, 2019
<i>International Workshop on Computer Vision & Intelligent Systems (IWCVIS)</i> | ◦ Merit Based Scholarship 2016 - 2020
<i>BRAC University, Dhaka</i>
70% tuition fee waiver |
| ◦ Champion - Programming Contest 2018
<i>BRAC University, Dhaka</i>
Inter Department Programming Contest | ◦ Certificate of Excellence (National) SSC '13 & HSC '15
<i>Prothom Alo, Dhaka</i>
Country's Highest Achievers Award |